

AC2205 Marking Scheme

Intermediate Management Accounting · Summer 2022 · Paper 1

Examination Session and Year	Summer 2022
Module Code	AC2205
Module Title	Intermediate Management Accounting
Paper Number	1
External Examiner	Prof Ivo De Loo
Head of Department	Prof Mark Mulcahy
Internal Examiner	Dr Peter Cleary
Duration of Paper	1.5 hours
Special Requirements	Students are allowed to use calculators that have standard arithmetic, scientific, graphical or algebraic functions provided that they are NOT programmable and can NOT store text (other than algebraic notation). Show all your workings.

Instructions to Candidates (as per exam paper): Answer two questions only. All questions carry equal marks (50 each). Show all your workings.

Own-figure rule: where a candidate makes an error at an early stage of a multi-step numerical question, full method marks are awarded for correctly applying the required technique to their own (incorrect) figure in all subsequent steps. Marks are never cascaded away for a single early error.

Marking philosophy: this is a closed-book, time-pressured examination. Academic citations are never required. Discursive answers are marked on depth of explanation and application to the specific scenario, not generic textbook recall.

Question Overview

Question	Scenario	Topics	Marks
Question 1	Smart Ltd	Flexible budgeting, flexible budget and volume variances, performance evaluation, criticisms of budgeting	50
Question 2	Kirk Ltd	Standard costing variance analysis (sales, material, labour, fixed overhead), profit reconciliation, problems with standard costing	50
Question 3	Sink Ltd	Cash budgeting with a minimum balance requirement and short-term borrowing/repayment mechanics	50

Candidates answer any TWO of the three questions above; each is worth 50 marks. This marking scheme covers all three questions in full.

QUESTION 1**Smart Ltd**

Total: 50 marks

Smart Ltd manufactures the "Clone" smart-speaker. The Managing Director, Nora Brown, wants to assess 2021 trading performance, comparing budget to actual, where actual production and sales were 10% lower than the original 15,000-unit plan due to supply chain issues. Candidates are required to (a) construct a flexible budget and calculate flexible budget variances and volume variances, (b) evaluate Smart Ltd's financial performance based on that analysis, and (c) outline four criticisms of budgeting.

Part	Requirement	Marks
(a)	Flexible budget, flexible budget variances (FBV) and volume variances (VV), each marked favourable or unfavourable	24
(b)	Evaluate 2021 financial performance and advise Ms. Brown	14
(c)	Four criticisms of budgeting	12

Question 1(a)**24 marks**

Smart Ltd, 2021, "Clone" smart-speaker: Budget (based on 15,000 units) and Actual figures given for sales revenue, variable costs (direct materials, direct labour, production overheads, marketing, distribution) and fixed costs (insurance, salaries, rent, utilities, other). Actual production and sales were 10% lower than the 15,000-unit plan. (a) Construct a flexible budget for Smart Ltd for 2021. Then, using the flexible budget, master budget and actual amounts, calculate flexible budget variances (FBVs) and volume variances (VVs) for each budgetary item, indicating whether each variance should be considered favourable (F) or unfavourable (U).

Full Model Answer**Working - actual volume and per-unit budget rates**

Item	Value
Master budget volume	15,000 units
Actual volume (10% lower than plan)	13,500 units
Budgeted selling price per unit ($€1,425,000 \div 15,000$)	€95.00

Method: The flexible budget re-states the master budget's variable items at the *actual* volume of 13,500 units, using the *budgeted* rate per unit for each; fixed costs remain unchanged in the flexible budget.

Full Model Answer (continued) – Flexible budget and variances: variable items**Flexible budget, FBV and VV – sales revenue and variable costs**

Item	Master budget (€)	Flexible budget (€)	Actual (€)	FBV (Actual – Flex)	VV (Flex – Master)
Sales revenue	1,425,000	1,282,500	1,248,750	(33,750) U	(142,500) U
Direct materials	450,000	405,000	415,000	(10,000) U	(45,000) F
Direct labour	337,500	303,750	291,500	12,250 F	(33,750) F
Production overheads	161,250	145,125	150,000	(4,875) U	(16,125) F
Marketing	71,250	64,125	66,800	(2,675) U	(7,125) F
Distribution	54,000	48,600	47,100	1,500 F	(5,400) F

Sign convention for costs vs revenue: For sales revenue, a shortfall against the comparison figure is unfavourable. For costs, a **lower** actual/flexible figure is favourable, even where the raw arithmetic is negative — each variance is labelled F or U according to its economic meaning, not the arithmetic sign.

Contribution

Item	Master budget (€)	Flexible budget (€)	Actual (€)	FBV	VV
Contribution	351,000	315,900	278,350	(37,550) U	(35,100) U

Full Model Answer (continued) – Fixed costs and net profit**Flexible budget, FBV and VV – fixed costs**

Item	Master budget (€)	Flexible budget (€)	Actual (€)	FBV (Actual – Flex)	VV (Flex – Master)
Insurance	35,000	35,000	50,000	(15,000) U	0
Salaries	100,000	100,000	120,000	(20,000) U	0
Rent	20,000	20,000	18,000	2,000 F	0
Utilities	15,000	15,000	17,500	(2,500) U	0
Other fixed costs	5,000	5,000	6,000	(1,000) U	0
Total fixed costs	175,000	175,000	211,500	(36,500) U	0

Net profit

Item	Master budget (€)	Flexible budget (€)	Actual (€)	FBV	VV
Net profit	176,000	140,900	66,850	(74,050) U	(35,100) U

Overall reconciliation: The total variance in net profit (Actual €66,850 – Master budget €176,000 = €(109,150) unfavourable) is fully explained by the sum of the flexible budget variance (€(74,050) unfavourable) and the volume variance (€(35,100) unfavourable): €(74,050) + €(35,100) = €(109,150).

Mark Allocation

Marks	Criteria
20–24	Actual volume correctly derived (13,500 units); flexible budget correctly constructed for all variable items; fixed costs correctly held constant; all FBVs and VVs correctly calculated and correctly labelled F/U; net profit reconciliation correct.
14–19	Substantially correct with minor errors in one or two line items, or F/U labelling inconsistently applied.
7–13	Flexible budget concept understood but significant errors in the FBV/VV split, or fixed costs incorrectly flexed with volume.
0–6	Limited understanding of flexible budgeting; minimal correct workings.

Question 1(b)**14 marks**

Based upon your answer to (a), briefly evaluate Smart Ltd.'s financial performance during 2021 in relation to "Clone", offering whatever advice to Ms. Brown that you consider appropriate.

Full Model Answer**Headline result**

Actual net profit of €66,850 fell €109,150 (62%) short of the €176,000 master budget — a very substantial proportional shortfall, even though the business remained profitable overall.

The volume variance (€(35,100) unfavourable) reflects the supply chain disruption

This portion of the shortfall stems directly from the 10% fall in production and sales volume, described as an unexpected short-term supply chain issue, and does not on its own reflect poor operational management during the period.

The flexible budget variance (€(74,050) unfavourable) is the larger, more concerning component

This is roughly double the volume variance and represents genuine spending and efficiency problems *after* already adjusting for the lower volume achieved. Within the variable costs, direct labour and distribution were favourable, but direct materials, production overheads and marketing were all unfavourable at the actual volume. Most significantly, every single fixed cost line except rent overspent against budget, with **salaries** (€(20,000) unfavourable) and **insurance** (€(15,000) unfavourable) the two largest overruns, together accounting for the majority of the total fixed cost overspend.

Advice to Ms. Brown

Investigate the salary and insurance overruns as the priority, since these are the largest controllable drivers of the shortfall and together explain most of the fixed cost overspend of €36,500. Recognise that variable cost control, while mixed, was not the dominant issue. Separately, given that the volume shortfall alone accounts for roughly a third of the total profit variance, Smart Ltd should review the resilience of its supply chain arrangements to reduce the risk of a similar disruption recurring.

Mark Allocation

Marks	Criteria
12-14	Correctly separates the volume-driven shortfall from the larger flexible budget variance; identifies salaries and insurance as the dominant drivers of the unfavourable FBV; gives specific, evidence-based advice referencing the figures from part (a).
8-11	Good evaluation distinguishing FBV from VV with reasonable identification of key drivers; advice more generic.
4-7	Some relevant commentary but does not clearly distinguish the two variance types, or lacks specific reference to individual line items.
0-3	Minimal or no meaningful evaluation.

Question 1(c)**12 marks**

Outline and briefly explain FOUR criticisms associated with the use of budgeting that Ms. Brown should be aware of.

Full Model Answer**1. Time-consuming and costly to prepare**

Preparing a detailed annual budget can consume significant management time and organisational resources that could otherwise be spent on value-adding activities.

2. Rigid/inflexible once set, without a flexible budgeting adjustment

A fixed master budget quickly becomes an inappropriate basis for comparison once actual conditions change, as illustrated directly by this question: comparing Smart Ltd's actual results straight against the original 15,000-unit master budget would produce a misleading picture, requiring the flexible budget adjustment in part (a) to give a fair, like-for-like comparison.

3. Can encourage dysfunctional behaviour and budgetary slack

Where managers are evaluated against meeting budget targets, this can create an incentive to deliberately understate achievable revenue or overstate necessary costs when the budget is set, or to spend up to the full remaining budget near the end of a period purely to avoid a reduced allocation next year, regardless of genuine need.

4. Backward-looking and can perpetuate inefficiency

Particularly where an incremental budgeting approach is used, budgets are often built from the prior period's figures with minor adjustments, rather than being rigorously reassessed from first principles, risking existing inefficiencies simply being carried forward year after year and encouraging a short-term focus on meeting the current year's target at the expense of longer-term strategic objectives.

Other acceptable criticisms: Candidates may alternatively discuss: budgets can become outdated quickly in a fast-changing or uncertain trading environment (as illustrated by the unexpected supply chain disruption in this question); they may not clearly link departmental targets to overall organisational strategy or value creation; or they may create silo behaviour between departments competing for a limited budget allocation. Any four well-explained, distinct criticisms receive full credit.

Mark Allocation

Marks	Criteria
10-12	Four distinct, well-explained criticisms given, each a genuine and recognised limitation of budgeting; ideally at least one linked to the specific issues encountered in this question.
7-9	Four criticisms identified with reasonable explanation, or three explained in excellent depth.
4-6	Two to three criticisms identified with limited explanation.
0-3	One or no relevant criticisms identified.

QUESTION 1 TOTAL — 50 MARKS

QUESTION 2**Kirk Ltd**

Total: 50 marks

Kirk Ltd designs and sells the "Power" Hoover. The Managing Director, David Smith (a non-accountant), wants to understand the firm's 2021 performance using standard costing variance analysis. Candidates are required to (a) reconcile budgeted and actual profit through a full set of sales, cost, and fixed overhead variances, (b) critically evaluate 2021 performance, and (c) outline four problems associated with standard costing.

Part	Requirement	Marks
(a)	Full variance analysis reconciling budgeted profit to actual profit, with assumptions stated	28
(b)	Critical evaluation of 2021 performance and advice to Mr. Smith	10
(c)	Four problems associated with standard costing	12

Question 2(a)**28 marks**

Kirk Ltd, 2021 Budgeted: Sales price €615/unit; Material 6kg × €35/kg = €210/unit; Labour 3hrs × €25/hr = €75/unit; Total fixed overheads €215,000; Production and sales 8,000 units. Actual: Total sales revenue €4,640,000; Total material costs (36,250 kgs) €1,377,500; Total labour costs (25,375 hours) €583,625; Total fixed overheads €345,000; Production and sales 7,250 units. (a) Using variance analysis, reconcile the budgeted and actual profit (loss) figures for Kirk Ltd. for 2021. Show all your workings and state clearly any assumptions that you make.

Full Model Answer - Preliminary workings**Working - standard cost and budgeted profit per unit**

Item	Value
Standard material cost per unit (6 kg × €35)	€210.00
Standard labour cost per unit (3 hrs × €25)	€75.00
Standard fixed overhead absorption rate (€215,000 ÷ 8,000 units)	€26.875 per unit
Standard full cost per unit	€311.875
Standard selling price per unit	€615.00
Standard profit per unit	€303.125

Budgeted profit and actual profit

Item	Value
Budgeted profit (8,000 units × €303.125)	€2,425,000
Actual sales revenue	€4,640,000
Less: Actual material costs	(€1,377,500)
Less: Actual labour costs	(€583,625)
Less: Actual fixed overheads	(€345,000)
Actual profit	€2,333,875

Assumptions made

1. Absorption (full) costing is used, with fixed overhead absorbed into each unit at the standard rate of €26.875, derived from budgeted fixed overheads ÷ budgeted units (no separate machine/labour hour absorption base is given). **2.** All materials purchased during the period were also used in production during the period (i.e. the material price variance is calculated on the actual quantity used, since no separate purchase quantity is given and no opening/closing raw material stock is mentioned). **3.** There is no opening or closing finished goods inventory — production and sales volumes are identical in both the budget (8,000 units) and actual (7,250 units) figures, so no additional stock-related adjustment is required.

Full Model Answer (continued) – Sales variances**Sales Volume Variance**

Item	Value
(Actual units – Budgeted units) × Standard profit per unit	$(7,250 - 8,000) \times €303.125$
Sales Volume Variance	€(227,343.75) Unfavourable

Interpretation: Kirk Ltd sold 750 fewer units than budgeted, each of which would have contributed €303.125 of standard profit — an unfavourable variance of €227,343.75 attributable purely to the shortfall in volume.

Sales Price Variance

Item	Value
Actual selling price per unit ($€4,640,000 \div 7,250$)	€640.00
(Actual price – Standard price) × Actual units sold	$(€640.00 - €615.00) \times 7,250$
Sales Price Variance	€181,250 Favourable

Interpretation: Kirk Ltd sold each unit for €25 more than the standard price, generating a favourable variance of €181,250 on the 7,250 units actually sold.

Full Model Answer (continued) – Material variances**Working**

Item	Value
Actual price per kg ($€1,377,500 \div 36,250$ kg)	€38.00
Standard quantity of material allowed for actual output ($7,250$ units $\times 6$ kg)	43,500 kg

Material Price Variance

Item	Value
(Standard price – Actual price) $\times$ Actual quantity used	$(€35.00 - €38.00) \times 36,250$ kg
Material Price Variance	€(108,750) Unfavourable

Material Usage Variance

Item	Value
(Standard quantity for actual output – Actual quantity used) $\times$ Standard price	$(43,500 - 36,250$ kg) $\times$ €35.00
Material Usage Variance	€253,750 Favourable

Interpretation: Kirk Ltd paid €3 more per kilogram of material than standard (unfavourable), but used 7,250 kg less material than the standard allowance for the units actually produced (favourable) — a large, efficient reduction in material usage relative to output, more than offsetting the adverse price effect.

Full Model Answer (continued) – Labour variances**Working**

Item	Value
Actual rate per hour ($€583,625 \div 25,375$ hours)	€23.00
Standard hours allowed for actual output ($7,250$ units $\times 3$ hrs)	21,750 hours

Labour Rate Variance

Item	Value
(Standard rate – Actual rate) $\times$ Actual hours	$(€25.00 - €23.00) \times 25,375$ hrs
Labour Rate Variance	€50,750 Favourable

Labour Efficiency Variance

Item	Value
(Standard hours for actual output – Actual hours) $\times$ Standard rate	$(21,750 - 25,375$ hrs) $\times$ €25.00
Labour Efficiency Variance	€(90,625) Unfavourable

A possible trade-off worth flagging to Mr. Smith: Kirk Ltd paid €2 less per labour hour than standard (favourable) but took 3,625 more hours than the standard allowance for the units produced (unfavourable) — a pattern consistent with using less experienced or lower-cost labour that works less efficiently, a classic real-world trade-off between a favourable rate variance and an adverse efficiency variance that is explored further in part (c).

Full Model Answer (continued) – Fixed overhead variances and full reconciliation**Fixed Overhead Expenditure Variance**

Item	Value
Budgeted fixed overheads – Actual fixed overheads	€215,000 – €345,000
Fixed Overhead Expenditure Variance	€(130,000) Unfavourable

Fixed Overhead Volume Variance

Item	Value
(Actual units – Budgeted units) × Standard fixed OH rate per unit	(7,250 – 8,000) × €26.875
Fixed Overhead Volume Variance	€(20,156.25) Unfavourable

Interpretation: Kirk Ltd spent €130,000 more on fixed overheads than budgeted (expenditure variance), and additionally under-absorbed fixed overhead by €20,156.25 because actual production (7,250 units) fell short of the budgeted volume (8,000 units) used to set the absorption rate (volume variance).

Full reconciliation: budgeted profit to actual profit

Item	€
Budgeted profit	2,425,000.00
Sales Volume Variance	(227,343.75)
Standard profit on actual sales	2,197,656.25
Sales Price Variance	181,250.00
Material Price Variance	(108,750.00)
Material Usage Variance	253,750.00
Labour Rate Variance	50,750.00
Labour Efficiency Variance	(90,625.00)
Fixed Overhead Expenditure Variance	(130,000.00)
Fixed Overhead Volume Variance	(20,156.25)
Actual profit	2,333,875.00

Check: The reconciliation ties out exactly to the actual profit of €2,333,875 independently calculated in the preliminary workings, confirming all eight variances have been correctly calculated and signed.

Mark Allocation

Marks	Criteria
24-28	Budgeted and actual profit correctly established; all eight variances (sales price, sales volume, material price, material usage, labour rate, labour efficiency, FOH expenditure, FOH volume) correctly calculated with correct F/U direction; full reconciliation ties out exactly to actual profit; assumptions clearly stated.
17-23	Most variances correct with one or two computational errors; reconciliation close to, but not exactly, actual profit.
9-16	Core variance formulae understood but applied with several errors, or one or more variances (e.g. FOH volume) omitted entirely.
0-8	Limited understanding of standard costing variance analysis; minimal correct workings.

Question 2(b)**10 marks**

Based upon all of the information at your disposal and using your answers from part (a) above, critically evaluate the 2021 performance of Kirk Ltd., offering whatever advice to Mr. Smith that you consider appropriate.

Full Model Answer**Headline result: actual profit exceeded budget despite the volume shortfall**

Perhaps counter-intuitively, Kirk Ltd's actual profit of €2,333,875 was only marginally below the budgeted €2,425,000 (a shortfall of €91,125, under 4%), even though 750 fewer units were sold than planned. This is because several large favourable variances offset the unfavourable ones almost entirely.

Strong pricing and material efficiency performance

The favourable sales price variance (€181,250) shows Kirk Ltd was able to sell 'Power' hoovers at a significant premium to the standard price — a genuinely positive commercial result worth understanding and, if possible, sustaining. The favourable material usage variance (€253,750) indicates production used materials substantially more efficiently than the standard allowed, a strong operational result.

A concerning trade-off in labour and a real cost-control problem in overheads

The favourable labour rate variance (€50,750) paired with the much larger unfavourable labour efficiency variance (€(90,625)) suggests Kirk Ltd may have used cheaper, but less efficient, labour — a trade-off that, on balance, cost the company money and should be investigated (is this a deliberate short-term staffing decision, or unplanned?). Separately, the €130,000 unfavourable fixed overhead expenditure variance is the single largest adverse variance in the whole analysis and represents a genuine, uncompensated cost overrun with no offsetting benefit — this is the item most deserving of Mr. Smith's attention.

Advice to Mr. Smith

Prioritise investigating the €130,000 fixed overhead overspend, as this is the largest single adverse variance and, unlike the labour trade-off, has no favourable counterpart to explain or justify it. Review whether the labour rate/efficiency trade-off reflects a deliberate, worthwhile staffing strategy or an unplanned efficiency problem. Recognise that the material and pricing performance were genuine strengths in 2021 that should be preserved and, if possible, replicated going forward.

Mark Allocation

Marks	Criteria
8-10	Correctly identifies that actual profit was close to budget overall despite offsetting variances; specifically identifies the labour rate/efficiency trade-off and the FOH expenditure overrun as the key issues; gives specific, evidence-based advice referencing the figures from part (a).
5-7	Good evaluation identifying most key variances with reasonable explanation; advice more generic.
2-4	Some relevant commentary but does not engage specifically with the individual variances calculated in part (a).
0-1	Minimal or no meaningful evaluation.

Question 2(c)**12 marks**

Outline and briefly explain FOUR problems associated with the use of standard costing that Mr. Smith should be aware of.

Full Model Answer**1. Standards can quickly become outdated**

Standards are typically set once, in advance of the period, based on expected prices and efficiency levels at that time. In a changing business environment (input price inflation, new technology, supply market shifts), the standards can become unrealistic well before the period ends, reducing their usefulness for genuine cost control and making resulting variances less meaningful.

2. Can encourage dysfunctional trade-offs between variances

Because individual variances are often reported and reviewed separately, managers may be tempted to improve one variance at the expense of another — exactly the pattern potentially seen in Kirk Ltd's results in part (a), where a favourable labour rate variance (achieved perhaps by using cheaper labour) was accompanied by a larger unfavourable labour efficiency variance. Focusing on individual variances in isolation, rather than their combined effect, can lead to decisions that look good on one metric but harm overall performance.

3. Best suited to standardised, repetitive production

Standard costing works well where products and processes are stable and repeat consistently, allowing meaningful 'standard' quantities and prices to be set. It is much less useful for businesses with significant product customisation, short product life cycles, or modern flexible/just-in-time manufacturing environments, where a single 'standard' may not meaningfully apply.

4. Setting standards is time-consuming, costly and can be subjective

Establishing accurate, realistic standards for material prices, labour rates, and expected efficiency levels requires significant time, technical expertise and judgement, and is itself subject to estimation error and potential manipulation (e.g. building in slack to make standards easier to achieve).

Other acceptable problems: Candidates may alternatively discuss: variance investigation itself can be costly and time-consuming, and not every variance is material enough to warrant investigation; or that an excessive focus on meeting standard costs can discourage innovation and continuous improvement beyond the standard. Any four well-explained, distinct problems receive full credit.

Mark Allocation

Marks	Criteria
10-12	Four distinct, well-explained problems given, each a genuine and recognised limitation of standard costing; ideally at least one linked to the specific labour rate/efficiency trade-off observed in part (a).
7-9	Four problems identified with reasonable explanation, or three explained in excellent depth.
4-6	Two to three problems identified with limited explanation.
0-3	One or no relevant problems identified.

QUESTION 2 TOTAL — 50 MARKS

QUESTION 3**Sink Ltd**

Total: 50 marks

Sink Ltd manufactures bathroom sinks. Candidates are required to (a) prepare a cash budget for Quarter Four 2022, incorporating a €20,000 minimum monthly closing balance requirement met through short-term borrowing (with interest charged on repayment), and (b) analyse the resulting cash position with advice to management.

Part	Requirement	Marks
(a)	Cash budget for Q4 (Oct, Nov, Dec), incorporating the €20,000 minimum closing balance and borrowing/repayment mechanics	40
(b)	Analysis of the Q4 cash position, with advice to management	10

Question 3(a)**40 marks**

Sink Ltd, budgeted figures July–December 2022 (€): Sales, Purchases, Salaries, Marketing, Administration and Overheads given monthly. Credit sales (60% of sales) settled 60% in month of sale, 20% one month after, 20% two months after; cash sales (40%) receive a 5% discount. Credit purchases (50% of purchases) paid 70% in month of purchase, 10% one month after, 20% two months after. Salaries paid one month in arrears; marketing paid two months in arrears; overheads (including a €6,000 monthly depreciation charge) paid one month in arrears; administration paid as incurred. €55,000 investment income expected in October. New equipment purchased June 2022 for €410,000, paid in 10 equal monthly instalments from July 2022. €30,000 tax payment due in October. Old equipment sold for €30,000 in November and €55,000 in December. Cash balance at 30 September 2022 expected to be €17,800. From 1 October 2022, a minimum monthly closing cash balance of €20,000 applies, met by short-term borrowing (in €1,000 multiples) on the first day of the month, repaid on the last day of a later month, with 12% p.a. interest charged on repayment for the period outstanding. (a) Prepare a cash budget for Quarter Four 2022.

Full Model Answer - Working 1: cash receipts from sales**Cash receipts from sales**

Item	October (€)	November (€)	December (€)
Cash sales this month, net of 5% discount (40% of month's sales × 95%)	117,800	125,400	131,100
Credit sales settled in month of sale (60% of month's sales × 60%)	111,600	118,800	124,200
Credit sales settled one month after sale (prior month's credit sales × 20%)	34,200	37,200	39,600
Credit sales settled two months after sale (two months prior credit sales × 20%)	31,800	34,200	37,200
Total cash receipts from sales	295,400	315,600	332,100

Full Model Answer (continued) – Working 2: cash payments**Cash payments for purchases**

Item	October (€)	November (€)	December (€)
Cash purchases this month (50% of month's purchases, paid same month)	82,500	90,000	105,000
Credit purchases paid same month (50% of month's purchases × 70%)	57,750	63,000	73,500
Credit purchases paid one month after (× 10%)	7,250	8,250	9,000
Credit purchases paid two months after (× 20%)	13,500	14,500	16,500
Total cash payments for purchases	161,000	175,750	204,000

Salaries, marketing, overheads and administration

Item	October (€)	November (€)	December (€)
Salaries (one month in arrears: Sep, Oct, Nov paid)	36,000	35,000	38,000
Marketing (two months in arrears: Aug, Sep, Oct paid)	33,000	36,000	38,000
Overheads, excl. €6,000 depreciation, one month in arrears (Sep, Oct, Nov generated)	29,000	30,000	32,000
Administration (paid as incurred, same month)	23,000	25,000	30,000

Overheads note: The €6,000 monthly depreciation charge embedded in the overheads figure is a non-cash charge and is stripped out before applying the one-month payment lag: September overheads of €35,000 generate a cash payment of €35,000 – €6,000 = €29,000, paid in October, and so on.

Full Model Answer (continued) - Other cash items and balance before financing**Other cash items**

Item	October (€)	November (€)	December (€)
Equipment instalment (€410,000 ÷ 10, months 4, 5 and 6 of the schedule beginning July)	41,000	41,000	41,000
Tax payment	30,000	0	0
Investment income received (cash inflow)	55,000	0	0
Sale of old equipment (cash inflow)	0	30,000	0
Sale of old machine (cash inflow)	0	0	55,000

Cash budget before financing activity

Item	October (€)	November (€)	December (€)
Total receipts	350,400	345,600	387,100
Total payments	353,000	342,750	383,000
Net cash flow (before financing)	(2,600)	2,850	4,100
Opening cash balance	17,800	20,200	21,010
Balance before financing	15,200	23,050	25,110

Full Model Answer (continued) – Borrowing and repayment schedule

October – shortfall requires borrowing

The pre-financing balance of €15,200 is below the €20,000 minimum by €4,800. Since funds can only be borrowed in €1,000 multiples, Sink Ltd must borrow **€5,000**, the smallest multiple of €1,000 sufficient to reach the minimum (€4,000 would only give €19,200, still short). Closing October balance = €15,200 + €5,000 = **€20,200**. No interest is charged in the month of borrowing.

November – repayment must account for the interest cost, not just the principal

The pre-financing balance of €23,050 is €3,050 above the minimum. A naive approach might repay €3,000 of the €5,000 outstanding loan (the largest €1,000 multiple available within the surplus) — but this would leave only €23,050 – €3,000 – interest to close the month. Since the policy requires the *closing* balance (after all transactions, including interest) to be at least €20,000, the interest cost itself must be factored into how much can safely be repaid. Repaying the full €3,000 (outstanding 2 months, interest = €3,000 × 12% × 2/12 = €60) would leave a closing balance of €23,050 – €3,000 – €60 = €19,990 — **€10 below** the required minimum. Sink Ltd should therefore repay only **€2,000** instead: interest = €2,000 × 12% × 2/12 = €40, giving a closing balance of €23,050 – €2,000 – €40 = **€21,010**, which satisfies the €20,000 minimum. €3,000 of the loan remains outstanding.

Own-figure note on this judgement call: Candidates who instead repay the full €3,000 in November (on the basis that the interest is a small, unavoidable cost separate from the round-thousands repayment decision) receive own-figure credit for a defensible alternative interpretation, provided the resulting figures are carried through consistently — but the model answer above reflects a stricter, and arguably more correct, reading of the minimum *closing* balance requirement as stated.

December – remaining loan fully repaid

The pre-financing balance of €25,110 is comfortably above the minimum, with €3,000 of the original loan still outstanding (borrowed 1 October, so outstanding for 3 months by 31 December). Interest on repayment = €3,000 × 12% × 3/12 = €90. Repaying the full remaining €3,000 gives a closing balance of €25,110 – €3,000 – €90 = **€22,020**, comfortably above the €20,000 minimum, with the loan now fully cleared.

Full Model Answer (continued) – Cash budget for Quarter Four 2022**Cash budget for Quarter Four 2022, including financing activity**

Item	October (€)	November (€)	December (€)	Quarter total (€)
Balance before financing	15,200	23,050	25,110	
Add: Amount borrowed	5,000	0	0	5,000
Less: Amount repaid	0	(2,000)	(3,000)	(5,000)
Less: Interest paid on repayment	0	(40)	(90)	(130)
Closing cash balance	20,200	21,010	22,020	22,020

Quarter as a whole: Starting from the opening balance at 1 October (€17,800), the net operating cash flow across the quarter (€(2,600) + €2,850 + €4,100 = €4,350) plus the net effect of borrowing and fully repaying the short-term loan (€5,000 borrowed less €5,000 repaid less €130 total interest = €(130) net) gives €17,800 + €4,350 – €130 = €22,020, matching the December closing balance exactly.

Mark Allocation

Marks	Criteria
34-40	All receipts and payments correctly calculated and timed (including the depreciation adjustment and equipment instalment window); minimum balance requirement correctly identified as breached in October; borrowing correctly rounded up to €5,000; November repayment correctly capped at €2,000 to keep the post-interest closing balance compliant (or the €3,000 own-figure alternative applied consistently); December's remaining loan correctly cleared with 3 months' interest; all closing balances correct.
25-33	Substantially correct with minor timing or computational errors, most commonly in the repayment rounding or the interest period calculation.
13-24	Core cash budget structure correct but the minimum balance/borrowing mechanic misunderstood or omitted (e.g. borrowing not rounded to €1,000 multiples, interest not calculated on the correct outstanding period, or the interaction between repayment sizing and the minimum closing balance not recognised).
0-12	Limited understanding of cash budgeting; minimal correct workings.

Question 3(b)**10 marks**

Based upon your answer to (a) above, briefly analyse Sink Ltd.'s budgeted Quarter Four cash position, offering whatever advice to management that you consider appropriate.

Full Model Answer**Overall assessment**

Sink Ltd's underlying cash position is broadly stable and improving across the quarter: a modest October shortfall requiring a short-term loan of €5,000 is fully repaid by the end of December, with the closing balance rising steadily each month (€20,200 → €21,010 → €22,020). The minimum balance policy operates as intended, though the margin above the €20,000 threshold remains fairly tight throughout the quarter.

What is driving the position

October alone shows a small negative net operating cash flow (€(2,600)), largely due to the one-off €30,000 tax payment; November and December are both modestly positive even before the one-off investment income, and equipment sale proceeds provide additional support in October, November and December respectively.

Advice to management

1. The need to borrow is modest and short-lived, and the minimum balance policy is working as intended — management can be reasonably confident in the plan as it stands. **2.** Because the margin above the €20,000 minimum remains tight in October and November, management should treat the underlying sales and payment timing assumptions as important to monitor closely; a small adverse shift (e.g. slower credit sales collection) could quickly require additional borrowing. **3.** Once the loan is cleared in December, management should consider whether the €20,000 minimum balance policy remains the right level given the improving trend, or whether it should be revisited as trading conditions stabilise.

Mark Allocation

Marks	Criteria
9-10	Correctly identifies the improving trend and the modest, well-managed nature of the October borrowing; explains the key drivers (tax payment, investment income, equipment sales); gives specific, sensible advice referencing the figures in part (a).
6-8	Correctly identifies the overall position with reasonable explanation of drivers; advice more generic.
3-5	Some relevant commentary but lacks specific reference to the figures produced in part (a).
0-2	Minimal or no meaningful commentary.

QUESTION 3 TOTAL — 50 MARKS